Requirement	Indexer	SH
OS	64bit	64bit
CPU	(12-98 CPU core) x1 vCPU	(16 CPU core) x1 vCPU
RAM	12-128 GB	12 GB
N/W	1 GbEthernet	1GB Ethernet

Ports { 8089 → Splunkd Port or Management Port
8000 → Splunkweb → http://servername:8000
8191 → KVstore
9997 → Splunk listening Port

Splunk System Administration

Module 1 – Deploying Splunk

- Provide an overview of Splunk
- Identify Splunk Enterprise components → Indexer, SH, DS, LM, HF, Cluster Master, SHC
- Identify the types of Splunk deployments → ① Standalone ② Basic ③ Distributed (Searching → SH; Indexing + Parsing → Indexer; Input → UF; Forwarder Mgmt → DS)
- List the steps to install Splunk
- Use Splunk CLI commands
- Explore security best practices

→ ulimit: A safety mechanism to prevent single application from accidently crashing whole system. Mainly 2 limits –
① Open Files (ulimit -n) → max no of files or socket a Process can open
Indexer/SH → ulimit -n ≥ 64K
② User Processes (ulimit -u) → max no. of background Process/thread a user can open/create
Indexer/SH → ulimit -u ≥ 16K

ulimit -a

#Installation –
① tar -zxvf <splunk.tgz> -C /opt
② /opt/splunk/bin/splunk start --accept-license --answer-yes ↔ username: admin, password: xxxx
③ " " " /splunk enable boot-start -user <username> ← non-root user

⇒ compatible version of SH & SearchPeers(Indexer) – Must be same or higher version to Indexer

- 4 Stages: Input → Parse → Index → Search also calls (SearchPeers)
- 3 Tiers [① Collection Tier (UF, HF, InfoD) → ② Indexing-Tier (IndexCluster) → ③ Search-Tier (SHCluster)] + Management Layer [License Mgr, Dep. Ser., SHC Deployer, ClusterMgr] ← replicate Data; share KOs
- Do not run Splunk as Superuser/root/admin
- On window, splunk set to start automatically on boot. On Linux, add manually to boot-start
- How to change Splunk Server Name? → Login → Settings → Server Settings → General Settings → "Splunk server name" → Restart splunk

• Deployer = Splunk component distribute apps & config updates to SHCluster members. → "Pause Indexing if free disk space (MB) below" default 5000MB

Module 2 – Monitoring Splunk

- Use Splunk Health Report
- Enable the Monitoring Console (MC) → Can run on shared instance → ① With LicenseMaster ② " a Deployment Server (<50 Clients) ③ " IndexClusterManager node but with less load (less load than ①)
  → 2 Modes: ① Standalone ② distributed
- Use Splunk Assist
- Use Splunk Diag → Available from any instance

Assist: A cloud-connected service to leverage telemetry data for real-time insight

Module 3 – Licensing Splunk

2.1 Identify Splunk license types
2.2 Describe license violations
- Add and remove licenses

① forwarder license – set HF – No Indexing
② Free license – 500 MB/day – best No-alert, SS, clustering
③ Enterprise " – Full functionality
④ Enterprise Trial license – 60 days Full Fun d^n [Pre-selected in a brand-new installation]

⇒ License Pricing Type → Ingest Pricing (ingest pricing) / Workload Pricing (vCPU SVC)

⇒ License metering occurs at 'indexing phase'

different calculation but draw from same license quota

⇒ Event – Measured as Per data fullsize
Metric – 150 bytes per metric event
⇒ Daily License Quota Does Not Count → Summary Index + Internal logs + metadata, tsidx + Replicated data
⇒ UF do NOT require Enterprise License to be installed.
⇒ License Location → SPLUNK_HOME/etc/licenses
⇒ License Pooling → Single Pool (500GB) + Multiple Pools based on Indexer groups (200+200+100)
⇒ ① License Alert → indexing > daily Quota + Clears at midnight → ② License Warning → Alert, but license not increased by Midnight
③ License Violation { ① 5 warnings on Enterprise License in 30-day rolling period; ② 3 warnings on Free License } affect → ④ DOES NOT AFFECT Indexing or Searches; ⑤ Searches disabled if licenses/license/day with 45 warnings in 60 days

Governs Splunk functionality

Module 4 – Using Configuration Files

3.1 Describe Splunk configuration directory structure

⇒ .conf: [Stanza] attribute=value → stored in SPLUNK_HOME/etc
modify → CLI, RESTAPI, Web, direct editing
⇒ Documentation & example view → SPLUNK_HOME/etc/system/README
*.conf.spec & *.conf.example

⇒ If sourcetype is explicitly set up in inputs.conf. It can NOT be fine-tuned using [source] stanza value in props.conf

SearchHead → Props.conf = SearchTimeFields Extraction, lookups, etc.
Indexer → props.conf = Parsing, inputs.conf = what data is collected, which port to listen (9997) + HEC
Forwarder → outputs.conf = where to send data, props.conf = Limited Parsing, inputs.conf = what data to collected

⇒ Multiple same .conf files in different folders, BUT only one .conf exist in memory.
↳ Conflicts handled by based on Context ① Global Context [Index-Time] ② App/User Context

⇒ Precedence

① Index Time Precedence (Global Context) (A) In case of standalone — System local dir ≫ App local dir > App default dir > System default dir
(alphabetical A>B)

② Search Time Precedence (App/User Context) → Current user dir for app (local > default) > App dir - running app (local > default) > App dir - other app (local > default) > System dir (local > default)

3.2 Understand configuration layering process
3.4 Use btool to examine configuration settings
3.3 understand configuration precedence

(B) In case of Index Cluster → ① Peer-app local dir (Highest) ② system local ③ App local ④ Peer-app default ⑤ App default ⑥ System default (lowest)

Validate On disk config → btool CLI ⇒ splunk btool inputs list
" in-memory " → splunk show config OR REST API → splunk show config inputs

- btool scenario → what are the /var/log/secure.log input configurations & where are they specified?
  > splunk btool inputs list monitor://var/log/secure.log --debug
- ⇒ override config
  - Addition → Include [stanza] in local with new setting
  - Modification → Place existing in local with new value
  - Deletion → Place " " " " Non-value
- Force Reload → http://server:8089/debug/refresh
- Restart → splunk restart

Module 5 – Using Apps

- Describe Splunk apps and add-ons
- Install an app on a Splunk instance
- Manage app accessibility and permissions

- Building Splunk App-curve

Module 6 – Creating Indexes

- Learn how Splunk indexes function
- 4.2 Identify the types of index buckets
- Add and work with indexes
- Overview of metrics index

4.1 Describe index structure ↔ Event (unstructured) / Metric (structured) = 4 Things: Timestamp, metric name, value, dimension
4.1 check index data integrity → SHA256 based hashes on raw data slices + use indexer acknowledgement capability to prevent data loss
splunk check-integrity (index / bucket) → To ensure data has not been tampered for audit/legal purposes
4.4 Describe indexes.conf options
4.5 Describe fishbucket

⇒ Splunk indexes found in SPLUNK-DB (SPLUNK_HOME/var/lib/splunk/)
⇒ "_introspection" → To track system performance, Provide Performance data to monitoring console
Pre-configured Indexes:
- "_thefishbucket" → contain checkpoint information for file monitoring inputs (prevent duplicate indexing) → One Record Per monitored file
  File metadata: Head = Pointer to the file; Tail = Pointer for last left off indexing
- "summary" → for summary indexing of reports
- "main" → Default index for inputs, located in defaultdb directory
- _audit, _internal

→ Compression factor ~50%.

Module 7 – Managing Index

- Review Splunk Index Management basics
- 4.6 Identify data retention recommendations
- Identify backup recommendations
- Move and delete index data
- 9.5 Describe the use of the Fishbucket
- Restore a frozen bucket

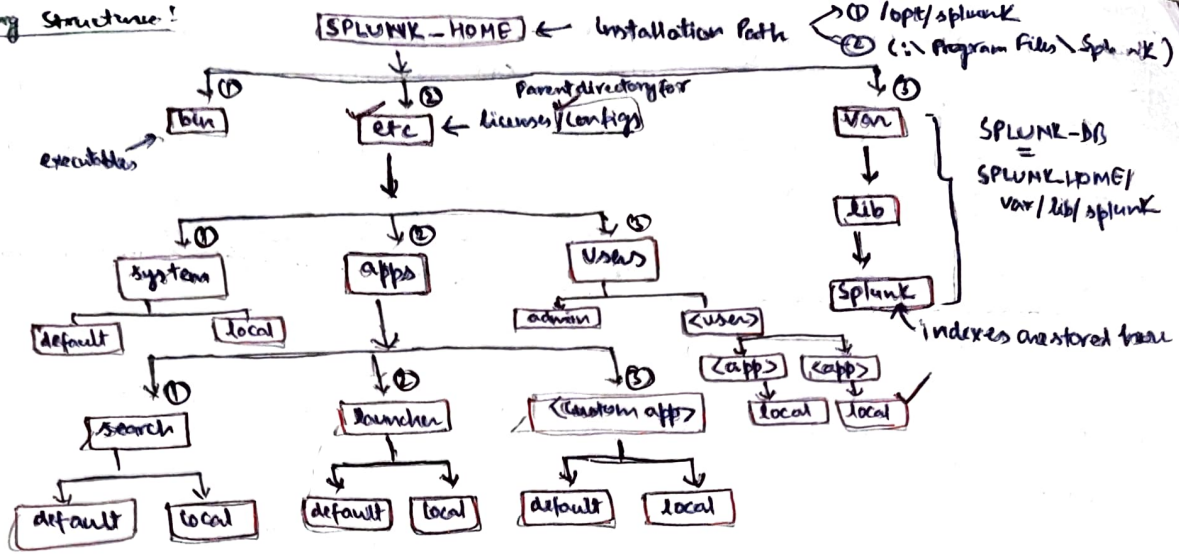
Apply a data retention Policy → Default 6 years; kept for x months → delete; " " " → archive
↳ 1 Retention Policy Per Index

→ Under indexes.conf (SPLUNK_HOME/etc/apps/<appname>/local), homePath, coldPath, thawedPath (3 Paths) MUST be specified, even when using default
→ Hot Buckets cannot be backup without stopping Splunk.

delete command to hide unwanted events from searches → kind of "Virtual deletion" by marking "deleted" → Hence No effect on disk space + CANNOT boundore + Need can_delete role

splunk clean → to delete ALL events from the index
(!) always specify "indexname", otherwise will flush ALL indexes.

⇒ Splunk Directory Structure:



configs { default → Overwritten on Update + DO NOT Modify
local → Preserved " " + Only modify This + Overrides default settings

#Splunk Server Name:
SPLUNK_HOME/etc/system/local/server.conf

```
[general]
serverName = ufo1
```

Default Host Name:
SPLUNK_HOME/etc/system/local/inputs.conf

```
[default]
host = ufo1
```

Index Creation ← Web / CLI
SPLUNK_HOME/etc/apps/<app_name>/local/indexes.conf

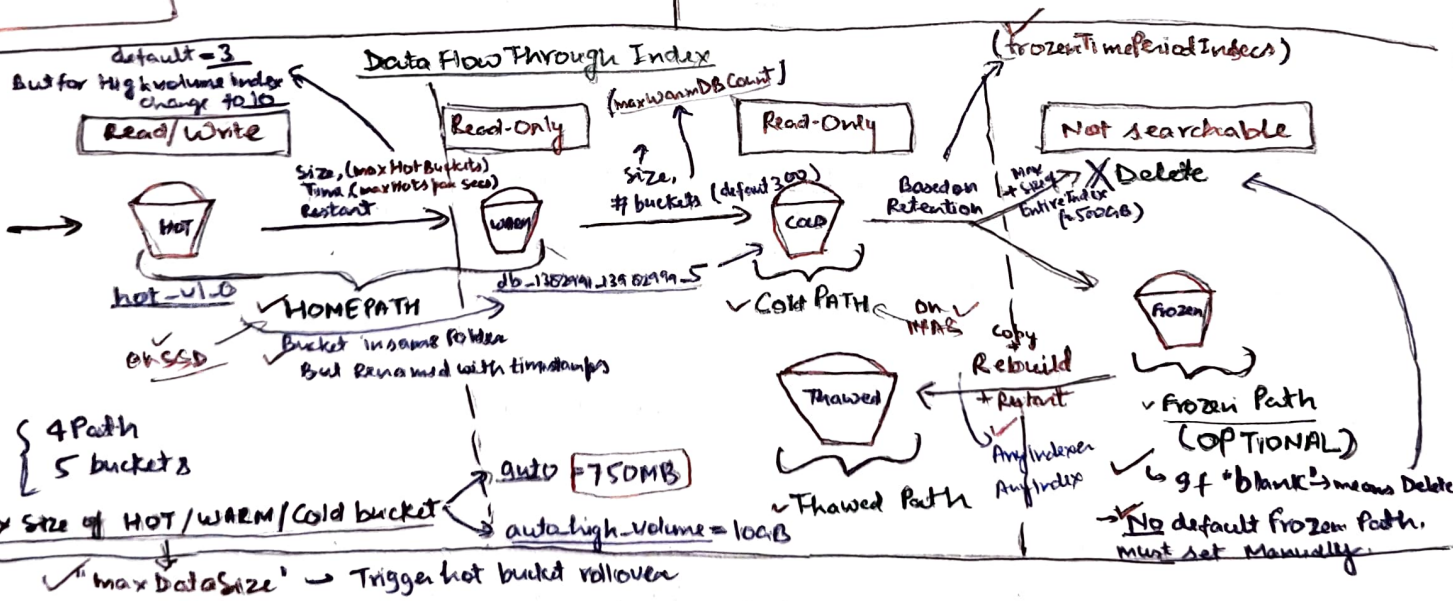
```
[test]
coldPath =
.
.
```

↑ If appname not defined in CLI → default app is search

Receiving Port ← Web → Setting > Forwarding and Receiving > Receive Data > New Receiving Port / CLI
SPLUNK_HOME/etc/apps/<app_name>/local/inputs.conf

```
[splunktcp://9997]
connection_host = ip
```

Data Flow Through Index



Module 8 – Managing Users

- 5.3 Add Splunk users using native authentication
- 5.1 Describe user roles in Splunk
- 5.2 Create a custom role
- Manage users in Splunk

Role = capabilities + Data(index) access
- ① Inherited from existing role [cannot be removed]
- ② selected capabilities

Unlock a user –
① Settings > Users > Edit > Unlock
② splunk edit user <locked-user> -locked-out false

⇒ Splunk stores native account passwords in SPLUNK_HOME/etc/passwd
→ Best Practice: Keep a failsafe account in passwd with strong password
To disable native authentication in splunk-create an empty passwd file

[for native-user: can be edited/deleted + Password Policy only applied here
for LDAP/non-native user – Only Time zone/default app can be changed]

⇒ Roles are in authorize.conf

DO NOT edit – SPLUNK_HOME/etc/system/default/authorize.conf
edit – SPLUNK_HOME/etc/system/local
edit – " " /apps/<app_name>/local

[• edit_user + edit_roles → allows user to promote themselves as full-admin
• edit_user + edit_roles_grantable → only allow to assign subset of role they have]

⇒ Authentication → LDAP, SAML + Scripted access to PAM, RADIUS
[LDAP group → Splunk Roles] authorize.conf [groupMapping] stanza

Module 9 – Configuring Basic Forwarding

- 7.1 Identify forwarder configuration steps
- 7.2 List splunk forwarder type → UF / HF (~Regular forwarder) / Light ✗ → deprecated after Splunk 6.6
- 7.3 Configure a Universal Forwarder
- 7.4 Add an inputs to UF using CLI
- Understand the Deployment Server

cancel configure → Receiving Ports

• UF bandwidth constraint 256 KBps (default)
UF [inputs.conf + outputs.conf] → Indexer [inputs.conf] ← Receiving Port 9997

⇒ on UF → splunk add forward-server <indexer: receiving-port> → modify outputs.conf stanza [tcpout-server ://<indexer:9997]
splunk add monitor /opt/file.log -sourcetype xyz -index test → modify inputs.conf [monitor: //opt/file.log]

only for one time upload of a file → splunk add oneshot <file with path> <index name>

⇒ Deployment server → ① Manage conf packages as 'apps' for 'clients' ② Forwarder Mgmt CLI ③ Listen on Mgmt Port 8089
④ Can Restart Remote Splunk instance ⑤ Requires Enterprise License ⑥ Should be on dedicated server.

⇒ By default forwarder (UF, HF) sends cooked data → To send raw data instead, in outputs.conf
→ for successful connection b/w UF & Indexer – check index=_internal host=<UFsn> | sendCookedData=false

Module 10 – Configuring Distributed Search

- 8.1 Describe how distributed search works
- 8.2 Define the roles of the search head and search peers
- 8.3 configure a distributed search group
- 8.4 list search head scaling options

8089

⇒ For Distributed search, add a user to all Indexers with "edit_role" capability + Indexer SH with license mgr + same index on all Indexer
⇒ Knowledge Bundles ← on SH → SPLUNK_HOME/var/run
on Indexer → SPLUNK_HOME/var/run/searchpeers

⇒ Search Peer Quarantine → Prevent Indexer from participating in future searches, if it is facing some issue
→ web → Settings > Distributed Searches > Search Peers
→ CLI → splunk edit search-server -auth <u:p> <host:port> -action quarantine

⇒ User logon to SH & run report ⇒ SH dispatch the searches to indexers (search peers) ⇒ Indexer run searches in parallel & return their portion of results ⇒ SH consolidate individual results & prepare report

⇒ SH → Dedicated → Doesn't share KO
→ Clustered → Shares a common set of KO

LSH = 8–12 simultaneous searches (adhoc + scheduled)
depends on CPU core on SH

⇒ Federated Search: conduct search & join data from local & Remote

How to enable –
on RSH → splunk user with fsh_manage capability
on FSH → configure RSH with account + Run SPL as –
index = federated:<indexname>

6.1 ☐ Integrate Splunk with LDAP ↔ LDAP = 389, Secure LDAP (LDAPS) = 636 → Splunk will search each LDAP strategy in the order they are listed in authentication.conf

6.2 ☐ List other user authentication options

6.3 ☐ Describe the steps to enable MFA in Splunk → Duo Security / RSA

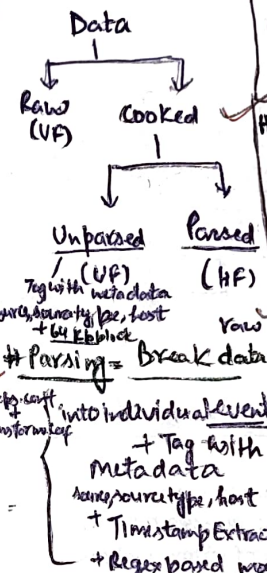
① "Login Page" appears → ② Enter credentials for authentication → ③ Authentication Granted → ④ Duo MFA → ⑤ Create User Session → ⑥ Log in to Splunk

⇒ Splunk checks internal database for a user login → then external like LDAP

⇒ What will happen to Login if MFA not working? ⇒ failOpen = true then Splunk bypasses Duo MFA when service is unavailable

↳ what will happen to logged in users if LDAP to Splunk role mapping changed? → Already logged in users will continue to operate till there next logout & login attempt

Feature & Capabilities	UF	HF
License	NO dedicated executable	Yes (Enterprise Forwarder License)
Bundle Python	NO	Yes
Fwd to 3rd Party	Yes	Yes
Pre-Event Filtering	NO	Yes
Event Routing	NO	Yes
Event Parsing	Sometimes	Yes
Local Indexing	NO	Yes, Optional
Splunk web	NO	optional
Searching / Alerting	NO	optional
Indexer Ack	Yes	Yes



Options for Custom Role
① Restrict search term (No whitelist)
② Limit the number of concurrent search jobs
③ Allow or Restrict the indexes

Power User
→ Create & share KO
→ NOT allowed to create Custom roles

MFA will work with only –
① Native authentication ✓
② LDAP ✓
③ Scripted authentication ✓
✗ SAML ✗ SSO

⇒ If want to reindex a file after some modification ⇒ fishbucket must be cleared or Reset at Forwarder itself

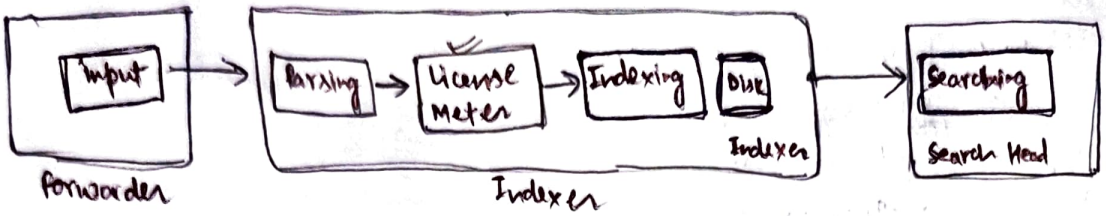
⇒ Distributed Search group = Group of search Peers
↳ specified in distsearch.conf file

```
[distributed Search: Paris]
default = false
servers = 192.168.1.1:8089, 192.168.1.2:8089
```

↳ use as –

```
index="..." , source="..." splunk_server_group=Paris
```

↳ NOT Valid for index clustering



Splunk Data Administration

Module 1 – Getting Data Into Splunk

- Provide an overview of Splunk
- Describe the Splunk distributed model
- Describe data input types and metadata settings
- Configure initial input testing with Splunk Web
- Testing Indexes with input staging

Module 2 –Config Files and Apps

- Identify Splunk configuration files and directories
- Describe index-time and search-time precedence
- Validate and update configuration files
- Explore Splunk apps and app installation

Module 3 – Configuring Forwarders

10.1 ☑ Configure Universal Forwarders

10.) ☑ Configure Heavy Forwarders

⇒ Selective Forwarding by UF ⇒ UF can route data to diff indexer servers based on source.

outputs.conf

```
[tcpout: SRVA]
server = SRVA:9997
[tcpout: SRVB]
server = SRVB:9997
```

⇒ To optimize HF

① Disable Indexing data on HF ⇒ outputs.conf

```
[IndexAndForward]
index = false
```

② Disable Splunk web on HF ⇒ web.conf

```
[settings]
startwebserver = 0
```

inputs.conf

```
[monitor: //.../metric.log]
_TCP_ROUTING = SRVA
[monitor: //.../event.log]
_TCP_ROUTING = SRVB
```

to selectively forward data to specific indexers or group of indexers.

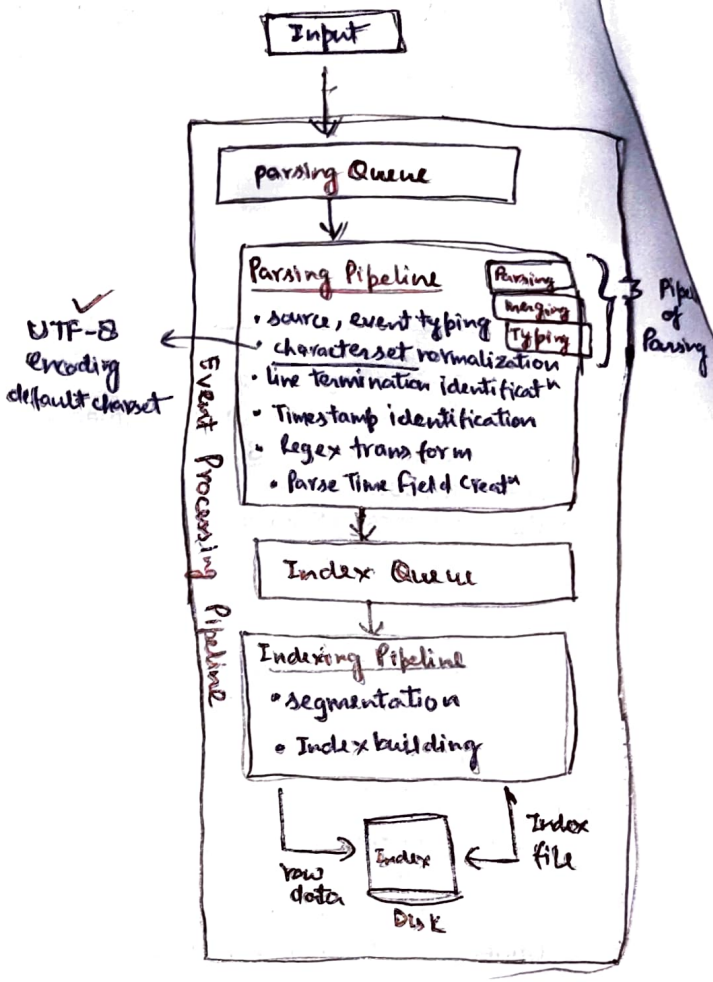
9.1 List 3 Phases of Splunk Indexing process

9.2 List Splunk input options

⇒ Inputs can be added -

1. Web
2. CLI
3. Apps & Add-on
4. inputs.conf

- File & Directories
- Network data
- Script output
- Linux & Windows logs
- HTTP
- ⋮



Module 4 – Customizing Forwarders

10.1 Configure intermediate forwarders → Configure Receiving Port on Intermediate Fwd → Splunk Web (if HF); splunk enable listen <fwd-port>; inputs.conf on Int. Fwd. [splunktcp://<fwdport>]

10.2 Identify additional forwarder options

11.1 Explain the use of deployment Management

Module 5 – Managing Forwarders

11.2 Describe Splunk Deployment Server (DS)

11.3 Manage forwarders using deployment apps

11.4/11.5 Configure deployment clients and client groups

11.6 Monitor forwarder management activities

Deployment Server

↳ To centrally manage configuration packages called "apps" for "clients"
↳ "Forwarder Management" as GUI → blacklist > whitelist
↳ can restart remote splunk instance
↳ listen from "clients/UF" on its Management Port 8089 (not 9997)

3 Components of Deployment Server

① Deployment-Apps: Config files packaged as "apps" at SPLUNK_HOME/etc/deployment-apps/ on DS server

② Server Classes: → Grouping of deployment clients based on IP/DNS name / Client name / Host name / *Machine-Type
↳ Define what app to deploy on which client
↳ is saved in serverclass.conf on DS at SPLUNK_HOME/etc/system/local/serverclass.conf

③ Deployment Clients: UFs which are connected to DS

App-Repository on DS: SPLUNK_HOME/etc/deployment-apps/<app>

deploymentclient.conf → exist on UF at SPLUNK_HOME/etc/system/local/deploymentclient.conf
↳ store which DS this UF should connect.
This is called Phone Home

Module 6 – Monitor Inputs

12.1 Create file and directory monitor inputs

12.2 Use optional settings for monitor inputs

12.3 Deploy a remote monitor input

- supports → CSV, XML, JSON, plaintext, multiple log4j + file compressed with gzip
- monitor new content using Fishbucket to keep checkpoint
- inputs.conf → [monitor://...] stanza
  ↳ default host → defined in SPLUNK_HOME/etc/system/local/inputs.conf [Default index is main]
  ↳ omitting sourcetype → splunk determine automatically
- ... (ellipsis) → recurses through directory & subdirectory (wildcard)
- * (asterisk) → matches anything in that specific directory

deployment client.conf on UF

[target-broker:deploymentServer]
targetUri = splunkDS_server:8089
PhoneHomeIntervalInSec = <int>

↓ interval to check DS for app update

Module 7 – Network Inputs

13.1 Create network (TCP and UDP) inputs

13.2 Describe optional settings for network inputs

→ If splunk restarts, data in transit may lost & not indexed as UDP do not check/resend.

- Network input on forwarder minimize indexer workload
- network input stanza in inputs.conf

[udp://<remotehost:port>]
connection_host = ip

[udp://:514]
connection_host = ip

[tcp://<remotehost:port>]
connection_host = dns

[tcp://:9002]
connection_host = dns

<remote_host> not specified means 2 things –
① All connection on this port will be matched
② minimum req. for N/W I/P stanza is protocol & port

connection_host attribute:
- dns = default for TCP → host is set to DNS name using reverse IP lookup
- IP = default for UDP → host is set to IP of originating host's IP
- none = Requires explicit setting for 'host' attribute

acceptfrom attribute:
↳ comma or space seperated list of addresses
↳ wildcard * can) ! (not)
↳ single IP / CIDR / DNS name

[udp://514]
connection_host = ip
acceptfrom = 10.1.1.6, 10.1.8

① Compress the feed by 2 ways

↳ ① Compress selected feed by setting on forwarder `outputs.conf` ⇒ compressed = true

↳ ② compress all feed " " " indexer → `inputs.conf` ⇒ " "

② Secure feed with SSL → `Encrypt → Compress → Reduce CPU utilization`

- splunk uses OpenSSL
- default certificate password is `password`

③ Automatic Load Balancing

① `outputs.conf` on UF/HF

```
[tcpout: my_LB_indexers]
server = idx1:9997, idx2:9997
```

↳ switch as per —

① AutoLBFrequency (default 30sec)

② autoLBVolume (default 0 = disabled)

✓ ③ when safe for data stream (EOF detected) even timing > 30sec

④ when receiving indexer goes down

④ Define Event Boundary in `props.conf` of UF

- Single line event

```
[my_syslog]
EVENT_BREAKER_ENABLE = true
```

- Multi line event

```
[my_log4j]
EVENT_BREAKER_ENABLE = true
EVENT_BREAK = ([\r\n]+)\d\d
```

⑤ Indexer Acknowledgement (disable by default)

⇒ outputs.conf

```
useACK = true
```

⑥ Forward Queue Size

↳ when forwarder cannot reach any indexer → data is queued in forwarder

`maxQueueSize = auto` without useACK → 500KB

⑦ Configure UF send data over HTTP

outputs.conf

```
[httpout]
httpEventCollectorToken = <authtoken>
uri = https://<ip>:8088
batchSize = 65536 (default 64KB)
batchTimeOut = 30 ( " 30sec)
```

To break events, on UF `props.conf`

```
LB_CHUNK_BREAKER = ([\r\n]+)
```

(default)

Override Host field

Method #1 constant 'host' name

Method #2 "host_segment" = 3 ⇒ 3rd segment of the directory path as hostname.

Method #3 "host_regex = <regex>" Extract host from path of file or filename

To reset file monitoring

① On source system for a specific file —

```
splunk cmd btprobe -d SPLUNK_DB/fishbucket/splunk_private_db
--file <source> --reset
```

② for all sources — rm -r SPLUNK_DB/fishbucket

↳ `splunk clean eventdata -index _fishbucket`

✓ followTail → ignore file's existing content, index new data

ignoreOlderThan = 6od ⇒ index event within last 6 days

✓ whitelist & blacklist ⇒ ✓ RegularExpression to filter files & folders (✓ blacklist > whitelist) in [monitor://] stanza of input file

✓ Note wildcards `...` & `*` are meant for stanzas while whitelist/blacklist are for attributes values

```
whitelist = \.log$
whitelist = query\.log$|my\.log$
```

↳ Pass both abc\query.log & abc\xyzquery.log

↳ Pass " abc\my.log & abc\xyzmy.log

```
whitelist = /query\.log$|/my\.log$
```

↳ Pass only abc/query.log, NOT abc/xyzquery.log

✓ 3 Network Input Queues [✓ Network Input, Not output]

① If Indexer can not be reached → data in Output Queue → attribute name "maxQueueSize" with default 500KB
② If "Output Queue" is full → data in "Memory Queue" → " " queueSize
③ " "memoryQueue" " " → " " "Persistent Queue" → " " "PersistentqueueSize" → does not exist by default
+ Preserved across restart as stored in Disk
+ written to SPLUNK_HOME/var/run/splunk/...

≠ Best Practice ① UDP = Splunk merge UDP data until it finds timestamp. Can be overridden during parsing phase
② Syslog = use syslog-collector like syslog-ng to write in file → then UF read → Prevent data loss due to its unavailability at UF during UF restart.
③ SNMP = Write trap to file → then use monitor input

Module 8 – Scripted Inputs → support .sh/.ps1/.bat/.py

13.3 ☐ Create a basic scripted input

✓ Splunk only execute scripts from 3 locations
① SPLUNK_HOME/etc/apps/<app-name>/bin
② SPLUNK_HOME/bin/scripts
③ " " /etc/system/bin
✓ To deploy the script to Production servers, copy script to SPLUNK_HOME/etc/deployment-apps/<app>/bin
→ Test script with "splunk cmd <scriptname>"
→ scripts inputs inherit Permission & User context of Splunkdaemon/ForwarderProcess

→ script stanza in inputs.conf
[script://<cmd>]
passAuth = <username> ⇒ To run script as a specified OS user
✓ interval = 60s(default) ⇒ or cronschedule } of 0 = Run script only at start up
✓ default "source" metadata = script identifier or path with prefix source::

✓ Script Input Buffering attributes
→ queueSize = Memory Queue
→ persistentQueueSize = Persistent Queue

Module 9 – Agentless Inputs

19.2 ☐ Configure Splunk HTTP Event Collector (HEC) agentless input
- Describe Splunk App for Stream

19.1 ☐ Create WMI inputs

→ HEC sends event to Splunk Event collector (Indexer / HF) without using forwarder using "Token based Authentication"
→ Secure + Scalable + send POST data in JSON format to "http[s]://<splunk-server>:8088/services/collector"
→ 3 options for HEC
① HEC Indexer Acknowledgment ⇒ ✓ Requires seperate channel provided by client
If enabled, after every HEC request, server returns Ack-identifier "ackID"
To check if data indexed, send request with array of ackIDs to "services/collector/ack" → server returns indexing status
② send "raw" payload to HEC → HEC allows any payload, not only JSON
③ Configure dedicated HTTP Setting ⇒ To limit who can access HEC endpoint – in server's inputs.conf

[http]
enableSSL = 1
Cross Origin Sharing Policy = *.splunk.com
acceptFrom = "!45.42.151/24, 157.73.224/24, *"

Module 10 – Operating System Inputs
- Identify Linux-specific inputs
- Identify Windows-specific inputs

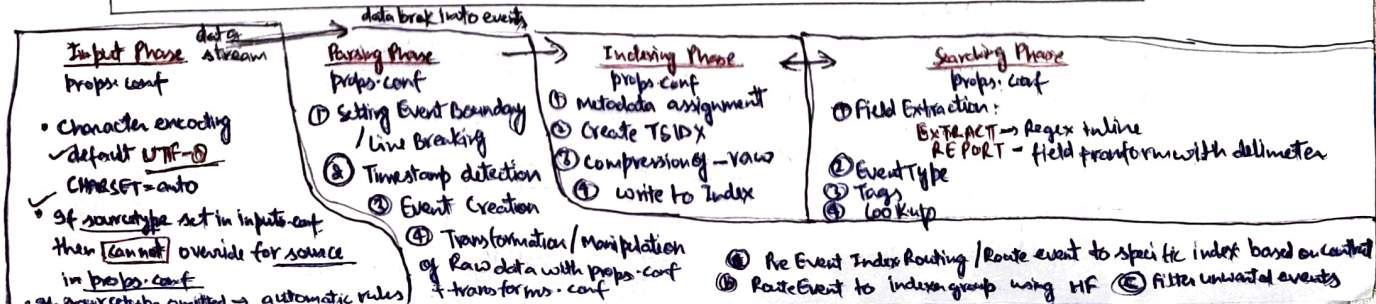
14.1 ☐

→ Splunk supports "journalctl" command for viewing logs collected by "systemd". Requires stanza in inputs.conf
→ Windows Eventlog input with whitelist/Blacklist [✓ Blacklist > whitelist] attributes in inputs.conf
→ ✓ Windows Remote Input with WMI for Eventlog, Performance } Remote inputs are configured in wmi.conf
↳ Advantage → collect input without forwarder
↳ Disadvantage → use WMI as transport protocol + Not recommended in high latency N/W + Requires splunk to run as domain account

Module 11 – Fine-tuning Inputs

15.1 ☐ Understand the default processing that occurs during input phase
15.2 ☐ Configure input phase options, such as source type fine-tuning and character set encoding

→ All data modification in props.conf are based on either source (1), sourcetype (2) or host (3) → can use wild card * + regex run
source & host stanza not in sourcetype



⇒ Windows can prevent a splunk forwarder from reading a open file. If files needs tobe read while they are being written to - **Monitor No Handle** attribute to be created in input stanza

⇒ Curl command to send multiple Events from HEC:

```
curl "https://splunkserver:8088/services/collector" \
-H "Authorization : Splunk <Token>" \
-d '{"event": "Hell World"}, {"event": "Hola Mundo"}, "nested": {"key1": "value"}}'
```

→ For Powershell, stanza in Inputs.conf

```
[Powershell://<name>]
script = <command>
schedule = <number | cron>
```

→ For Eventlog with whitelist / Blacklist

```
[WinEventLog://Security]
disabled = 0
whitelist1 = EventCode = /^[15].*$/ Type=/Error|Warning/
whitelist2 = TaskCategory = %^Log.*$%
blacklist = 540
```

→ In wmi.conf for EventLog

```
[WMI:remote-logs]
interval = 5
server = server1, server2, server3
event_log_file = Application, Security, System
```

→ wmi.conf for Performance

```
[WMI:remote-perfomon]
interval = 5
server = server1, server2
wql = select DatagramsPerSec
```

→ Props.conf

```
[source::<sourcename>]
attribute = value
[host::host-name]
attribute = value
[<sourcetype-name>]
attribute = value
```

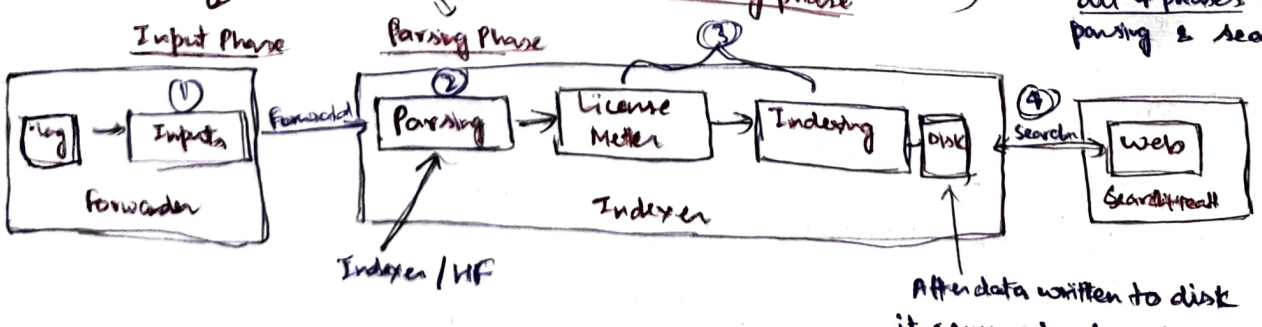
(without prefix "sourcetype")

```
[source::/var/log/secure*]
sourcetype = linux-secure
[host::myc*]
TZ = US/Eastern
[sendmail]
CHARSET = auto
```

[send*] ✗ → wildcard(*) & regex only allowed with host & source

⇒ Index-Time Process

→ Props.conf referenced during all 4 phases → inputs, indexing, parsing & searching



① Input Phase

↳ default metadata fields (source, sourcetype, host, index)

② Parsing Phase

↳ Parsing = break data stream into Events → each with Timestamp & Timezone

↳ Data-Stream → Breaks into event → Event-by-Event Processing

②·① Line Breaking / Event Boundary (= where an events starts and ends) + ✓ can be defined to certain limit at UF

```
SHOULD_LINEMERGE = false
LINE_BREAKER = <regex>
```
⇒ for single line event carriage return ([\r\n]+)

OR,
```
1 SHOULD_LINEMERGE = true (default)
2 LINE_BREAKER = ([\r\n]+)
3 (or BREAK_ONLY_BEFORE = <regex>
4 or BREAK_ONLY_BEFORE_DATE = true
5 or MUST_BREAK_AFTER = END)
```

②·② Timestamp detection [✓ can NOT be defined at UF]

↳ correct extraction is essential

→ Splunk makes its best attempt to identify event boundary & timestamp, however if familiar, provide more info to splunk through arguments

[my custom source & sourcetype]
1 TIME_PREFIX= ^ (to tell splunk where timestamp starts, like [\d+]\s+ for [12345] 2026-10-07 14:14:14
2 TIME_FORMAT = <strptime format> like %Y-%m-%d %H:%M:%S
3 MAX_TIMESTAMP_LOOKAHEAD = <int> 20 [How far splunk should search, ignore if latter timestamp in data]
4 DATETIME_CONFIG = CURRENT / NONE (special timestamp parser)
5 TZ = America/New York (used for timezone offset for correct time extraction under 'host/source/sourcetype' stanza of props.conf

✓ Steps of Splunk Timestamp Processing

① use TIME_FORMAT (props.conf) to identify timestamp in event

② If no TIME_FORMAT is configured, try automatically identify timestamp from events

②·① If identify "time + date", but NOT "year" : Determine a year

②·② " " "time", but NOT "date" : try to find timestamp in source name or file-name

②·③ If CANNOT identify "date" : use file modification time

③ Else NO timestamp found :

③·① If any timestamp from "same source", use most recent timestamp

③·② If NO timestamp, use "current system time while indexing!"

③ Event Creation

④ Transformations / Manipulating Raw data with props.conf + transforms.conf

④·① Mask or delete _raw data as it is being indexed —

2 methods of _raw data transformation —

① SEDCMD

↳ used only props.conf ✓

↳ only used to "mask" or "truncate" data (✓ NOT to eliminate), provide search & replace using regex

use case — hide first 5 digits of account in sales.log

props.conf —
```
[source::/.../sales.log]
SEDCMD-lacct = s/<find w/ regex>/<replace with>/1/g
```
[modify/mask _raw data before indexing]

② TRANSFORMS

↳ use props.conf + transforms.conf

↳ Pre-event-transformation → Transform matching events based on source/sourcetype/host →

Module 12 – Parsing Phase and Data Preview

16.1 ☐ Understand the default processing that occurs during parsing
16.2 ☐ Optimize and configure event line breaking
16.3 ☐ Explain how timestamps and time zones are extracted or assigned to events
16.4 ☐ Use Data Preview to validate event creation during parsing phase

↳ feature in splunk allows Event Breaking, Timestamp extractions and any advanced configuration found in props.conf to be validated through UI:

Module 13 – Manipulating Input Data

- Explore Splunk transformation methods
- Create rulesets with Ingest Actions
- Mask data with Ingest Action rules
- Mask data with SEDCMD and TRANSFORMS
- Override sourcetype or host based upon event values

12.1 ☐ Explain how data transformations are defined and invoked

12.2 ☐ Use transformation with props.conf & transforms.conf to
① Mask or delete raw data as it is being indexed
② Override sourcetype or host based upon event values
③ Route events to specific indexes based on event content
④ Prevent unwanted events from being indexed

Module 14 – Routing Input Data

- Filter data with Ingest Action rules
- Route data with Ingest Action rules
- Route data with TRANSFORMS

(4.2) Override sourcetype or host based upon event value using props.conf & transforms.conf

in props.conf

```
[sourcetype]
TRANSFORMS = <stanza name>
```

in transforms.conf

```
[<stanza name>]
1 SOURCE_KEY = <field name> (default = _raw)
2 REGEX = (event from SOURCE_KEY that will be processed)
3 DEST_KEY = (where to write processed data)
4 FORMAT = (Control how REGEX write to DEST_KEY)
```

PTO

USECASE#1: Mask sensitive data

props.conf

```
[source::...\store\purchase.log]
TRANSFORMS-1ccnum = c_num_anon
```

→ Lable "1ccnum" identifies this transforms "namespace"

transforms.conf (same stanza)

```
[c_num_anon]
REGEX = (.* cc_NUM:\s)\d{12} (\d{4}.*)
DEST_KEY = _raw
FORMAT = $1 xxxxxxxx$2
```

⇒ finds 2 capture groups (), referred in FORMAT as $1, $2

USECASE#2: Setting per event sourcetype

props.conf

```
[source::udp:514]
TRANSFORMS = custom_sourcetype
```

transforms.conf

```
[custom_sourcetype]
REGEX = Custom$
DEST_KEY = MetaData:Sourcetype
FORMAT = sourcetype::custom_log
```

⇒ If event ends with "Custom"

⇒ ("MetaData" key used for Host/Source/Sourcetype)

⇒ (when "MetaData" key used, then prefix in FORMAT is host::, source::, sourcetype::)

USECASE#3: Setting Per Event Hostname

props.conf

```
[sales_entries]
TRANSFORM-register = sales_host
```

transforms.conf

```
[sales_host]
REGEX = server:(\w+)
DEST_KEY = MetaData:Host
FORMAT = host::$1
```

⇒ If event has "server:<servername>")

(4.3) Pre Event Index Routing / Route events to specific indexes based on event content

props.conf

```
[mysrctype]
TRANSFORMS-itops = route_err_warn
```

transforms.conf

```
[route_err_warn]
REGEX = (Error|Warning)
DEST_KEY = _MetaData:Index
FORMAT = itops
```

(4.4) Route events to indexer's group using Hf

outputs.conf

```
[tcpout:syslogGroup]
server = server1:9997, server2:9997
```

props.conf

```
[syslog]
TRANSFORMS-routing = syslogRouting
```

transforms.conf

```
[syslogRouting]
REGEX = .
DEST_KEY = _TCP_ROUTING
FORMAT = syslogGroup
```

(4.5) Prevent / Filter Unwanted events from being index

↳ route to "nullQueue"

↳ discarded events will not be counted in license quota

props.conf

```
[WinEventLog:System]
TRANSFORMS = null_queue_filter
```

transforms.conf

```
[null_queue_filter]
REGEX = (?i)^EventCode=(592|593)
DEST_KEY = queue
FORMAT = nullQueue
```

⇒ (?i) means "ignore case"

⇒ (Route to "queue" & use "nullQueue" to discard events)

---

(5) License Meter → run just before Indexing Phase & "_raw" is metered for license usage

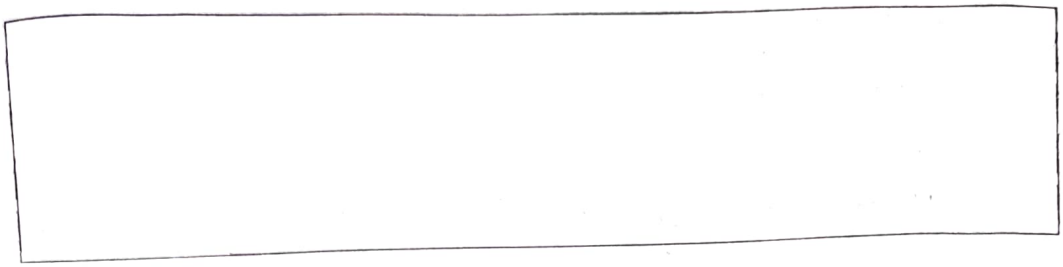
(6) Indexing Phase

- (6.1) Metadata assignment
- (6.2) Create TSIDX
- (6.3) Compression of _raw
- (6.4) write to Index (Disk)

→

Module 15 – Supporting Knowledge Objects

- Define default and custom search time field extractions
- Identify the pros and cons of indexed time field extractions
- Configure indexed field extractions
- Describe default search time extractions
- Manage orphaned knowledge objects



(7) Search Phase

(7.1) Field Extraction

- ✓ EXTRACT → in-line extraction based on 'Regex' saves as EXTRACT
- ✓ REPORT → fieldtransform based on 'Delimiter' " " REPORT

(7.2) Event Type

(7.3) Tags

(7.4) Lookups